Please Show Work and Provide Answers on Separate Paper

STAT 50 – Chapter 1 Quiz

1. A large population has a mean of 27 and a standard deviation of 3.2. A sample of size 50 is taken from the population. The sample mean is 26.6 and the sample standard deviation is 3.5. The values of the population mean and standard deviation are called ___parameters___, while the sample mean and sample standard deviation are called ___statistics___?

2. Describe how you would use a random number generator to collect a Simple Random Sample (SRS) of size 10 from a population of 100 individuals.

   Make one random number generator to assign an integer to everyone in the population between 1 and 100. Then, generatea list of random numbers between 1 and 100 and choose corresponding population individuals to make up the sample random sample.

3. Items in a Simple Random Sample may be treated as ***independent***, provided the sample size does not exceed ***5%*** of the population.

For questions 4 and 5, state whether the investigation is an observational study or a designed experiment. Explain your answer.

4. Researchers randomly assigned 39,876 initially healthy women 45 years of age or older to 100 mg of aspirin or placebo on alternative days and then monitored them for 10 years for a first major cardiovascular event (i.e. nonfatal myocardial infarction, non-fatal stroke, or death from cardiovascular causes). ***Designed experiment.***

5. A study involving 3800 people over the age of 65 in East Boston. The study took place over a six-year period and compared the age at which death occurred for right-handed people vs. left-handed people among the 3800 people who died during the six-year period. The study concluded there was no difference. ***Observational study.***

6. Cause and effect can sometimes be established in an ***designed experimental***, but generally not in an ***observational study***. Explain.

Because, in an experiment, independent variables will be designed in such a way that affects dependent variables. While observational study, just observes and records the outcomes.

7. Suppose a company that gives whale-watching tours wants to survey its customers. Out of ten tours they give one day, they randomly select four tours and ask every customer about their experience. What type of sampling is this?

Stratified random sample.

8. Suppose a high school principal wants to conduct a survey to collect the opinions of students. He first splits the students into four groups based on their grade – Freshman,

Sophomore, Junior, and Senior – then selects a simple random sample of 50 students from each grade to be included in the survey. What type of sampling is this?

Stratified random sample.

9. Suppose high school students are conducting a study on the average consumption of pizza in the cafeteria each week. They could call their classmates and ask how many slices they consume during the week. What type of sampling is this and why might it not be a good method?

This is an example of a sample of convenience, but it might not be good method because the population represents the whole high school, while sample uses only one class students, which is biased.

10. Malnutrition and Poverty. R. Reifen et. Al. studied various nutritional measures of Ethiopian school children and published their findings in the paper "Ethiopian-Born and Native Israeli School Children Have Different Growth Patterns" (Nutrition, Vol. 19, pp 427 – 431). The study, conducted in Azezo, North West Ethiopia found that malnutrition is prevalent in primary and secondary school children because of economic poverty. The weights, in kilograms (kg), of 60 randomly selected male Ethiopian-born school children aged 12 – 15 years old are presented in increasing order in the following table.

30.1	37.7	38.0	38.8	38.9	39.0
39.3	40.9	41.1	41.3	41.5	41.8
42.0	42.0	42.1	42.5	42.5	42.8
42.9	43.3	43.4	43.5	44.0	44.4
44.7	44.8	45.2	45.2	45.2	45.4
45.5	45.7	45.9	45.9	46.2	46.3
46.5	46.6	46.8	47.2	47.4	47.5
47.8	47.9	48.1	48.2	48.3	48.4
48.5	48.6	48.9	49.1	49.2	49.3
50.9	51.4	51.8	52.8	53.8	58.2

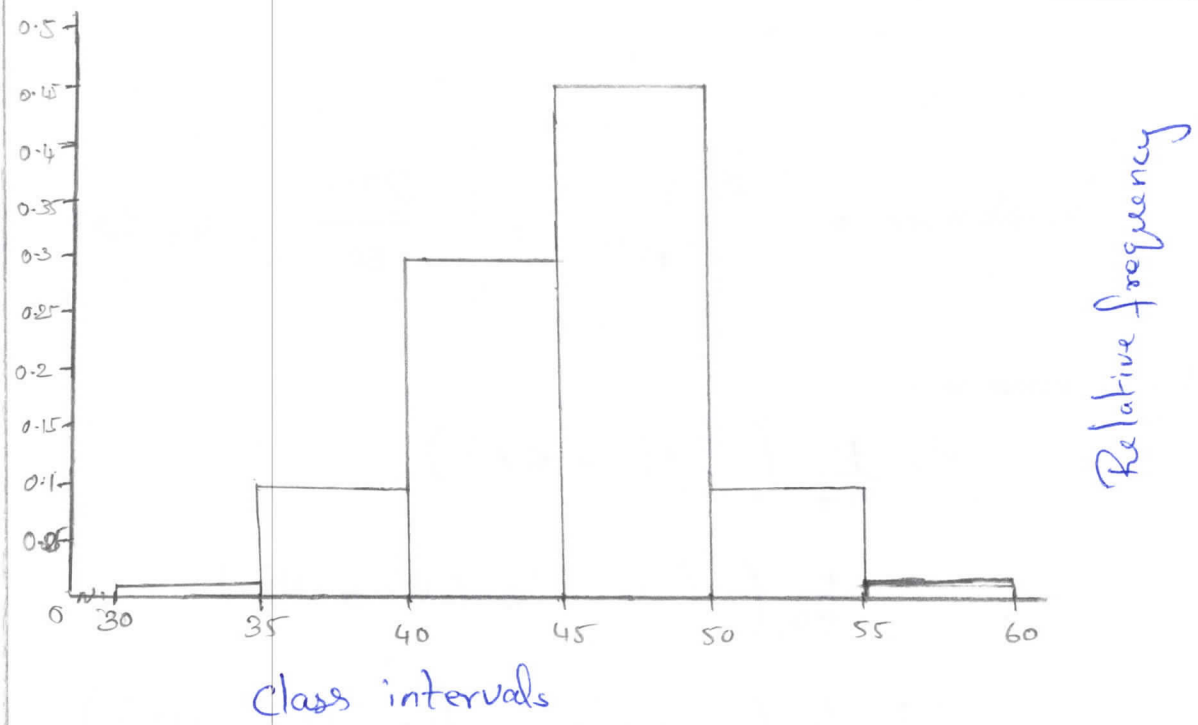
Hint: $\sum_{i=1}^{60} x_i = 2713$ and $\sum_{i=1}^{60} x_i^2 = 123{,}881.1$

(a) Construct a Relative Frequency Histogram for the data set. Use class intervals of width 5, with the first interval being 30 - <35, that is [30,35).
(b) Determine the five-number summary (round to three decimal places if necessary), then Construct a Box Plot for the data set. Be sure to show outliers, if any.
(c) Compute the sample mean for the data set.
(d) Compute the sample variance and sample standard deviation for the data set.
(e) Using your Histogram, comment on the characteristics of the data set. Specifically, comment on shape (roughly symmetric vs. skewed with respect to the mean not the median), center (mention two), and spread (mention two).

(10)

(a)

Class interval	Frequency	Relative frequency
[30, 35)	1	$1/60 = 0.017$
[35, 40)	6	$6/60 = 0.100$
[40, 45)	19	$\frac{19}{60} = 0.317$
[45, 50)	28	$\frac{28}{60} = 0.467$
[50, 55)	5	$\frac{5}{60} = 0.083$
[55, 60)	1	$1/60 = 0.017$



(b)

five-number summary :-

Min : 30.1

Max : 58.1

Q_1 : $0.25(60) = 15 = 42.1$

Q_2 : $0.50(60) = 30 = 45.4$

Q_3 : $0.75(60) = 45 = 48.1$

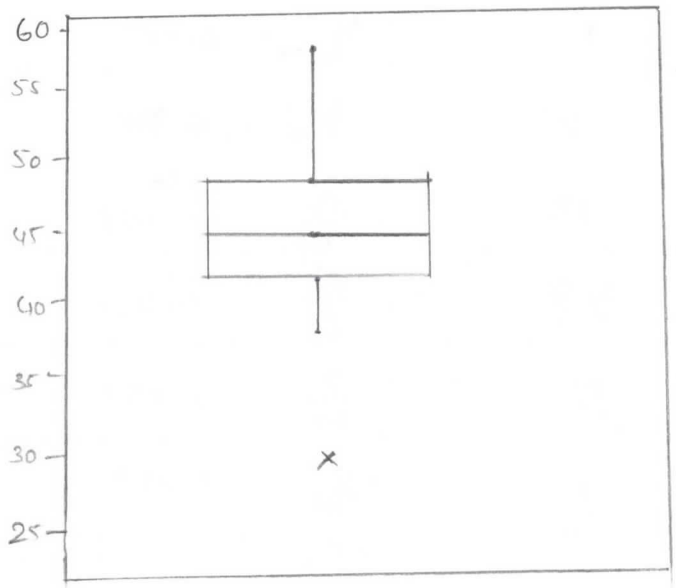
(b) check outliers:-

$IQR = Q_3 - Q_1 = 48.1 - 42.1 = 6$

Median = 45.4 , Then ⇒ low bound = 36.4

upper bound = 63.4

There is an outlier.



There is an outlier, which is 30.1.

(c) Mean:-

$$\text{Sample mean} = \frac{\sum_{i=1}^{60} x_i}{n} = \frac{2713}{60} = 45.217 \text{ kg}$$

(d) Sample variance:-

$$S^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)$$

$$= \frac{1}{59}\left(\sum_{i=1}^{60} x_i^2 - 60(45.217)^2\right)$$

$$= \frac{1}{59}\left(123881.1 - [60 \times (45.217)^2]\right)$$

$$S^2 = 20.44872 = 20.449$$

$\therefore$ $S^2 = 20.449$

Standard deviation:-

$$S = \sqrt{20.449} = 4.522026$$

$\therefore$ $S = 4.522$

(c) Shape: positively skewed, which is skewed to the right.

Center: the center is around 45.217kg mean and 45.4kg median.

Spread: The spread, IQR is 6kg and standard deviation is 4.522kg.